

Retail Sales Performance Analysis

Practice Questions & Solutions

Q1. Write a SQL query to retrieve all columns for sales made on '2022-11-05'

Solution:

```
SELECT
    *
FROM sales_data
WHERE sale_date IN ('2022-11-05');
```

Q2. Write a SQL query to retrieve all transactions where the category is 'Clothing' and the quantity sold is more than 2 in the month of Nov-2022.

Solution:

```
SELECT
    *
FROM sales_data
WHERE category IN ('Clothing')
    AND
    quantity > 2
    AND
    TO_CHAR(sale_date, 'YYYY-MM') = '2022-11'
```

Q3. Write a SQL query to calculate the total sales (total_sale) for each category.

Solution:

```
SELECT
    category,
    ROUND(SUM(total_sale) / 100000, 2) AS "sales in million"
FROM sales_data
GROUP BY 1
```

Q4. Write a SQL query to find the average age of customers who purchased items from the 'Beauty' category.

Solution:

```
SELECT
    FLOOR(AVG(age))
FROM sales_data
WHERE category IN ('Beauty');
```

Q5. Write a SQL query to find all transactions where the total_sale is greater than 1000.

Solution:

```
SELECT
    *
FROM sales_data
WHERE total_sale > 1000
ORDER BY 2 ASC;
```

Q6. Write a SQL query to find the total number of transactions (transaction_id) made by each gender in each category.

Solution:

```
SELECT
    category,
    COUNT (CASE WHEN gender = 'Male' THEN 1 END) AS male,
    COUNT (CASE WHEN gender = 'Female' THEN 1 END) AS female
FROM sales_data
GROUP BY 1;
```

Q7. Write a SQL query to calculate the average sale for each month. Find out best-selling month in each year

Solution:

```
SELECT
    TO_CHAR(sale_date, 'YYYY-MM') AS "Year-month",
    EXTRACT(YEAR FROM sale_date) AS year,
    LEFT(TO_CHAR(sale_date, 'Month'), 3) AS month,
    ROUND(AVG (total_sale), 2) AS "Average Monthly Sales"
FROM sales_data
GROUP BY 1, 2, 3
ORDER BY 1;
```

Q8. Write a SQL query to find the top 5 customers based on the highest total sales

Solution:

```
SELECT
    customer_id,
    ROUND(SUM(total_sale) / 1000 ,2) AS "Sales (in K)"
FROM sales_data
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

Q9. Write a SQL query to find the number of unique customers who purchased items from each category

Solution:

```
WITH unique_count AS
(
    SELECT
        DISTINCT
            customer_id,
            category
    FROM sales_data
)
SELECT
    category,
    COUNT(customer_id) AS unique_customer
FROM unique_count
GROUP BY 1
```

Q10. Write a SQL query to create each shift and number of orders (Example Morning <=12, Afternoon Between 12 & 17, Evening >17)

Solution:

```
SELECT
    CASE
        WHEN sale_time BETWEEN '06:00:00' AND '12:00:00' THEN
'Morning'
        WHEN sale_time BETWEEN '12:00:00' AND '16:00:00' THEN
'Afternoon'
```

```
        WHEN sale_time BETWEEN '16:00:00' AND '19:00:00' THEN
'Evening'
        WHEN sale_time BETWEEN '19:00:00' AND '24:00:00' THEN 'Night'
    END AS shift,
    COUNT(transactions_id) AS total transaction
FROM sales_data
GROUP BY 1
ORDER BY 1;
```